Adversarial attacks

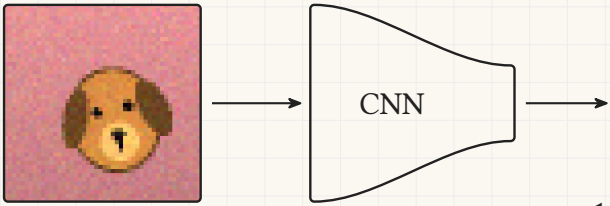
Weekend 4, Saturday morning

Carlos Cotrini

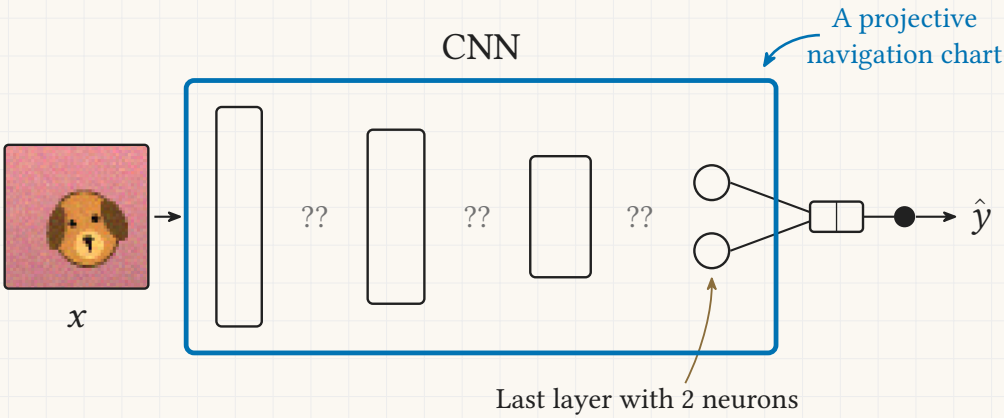
CAS Building ML/AI Applications, ETH Zurich

Saturday 24 October 2026

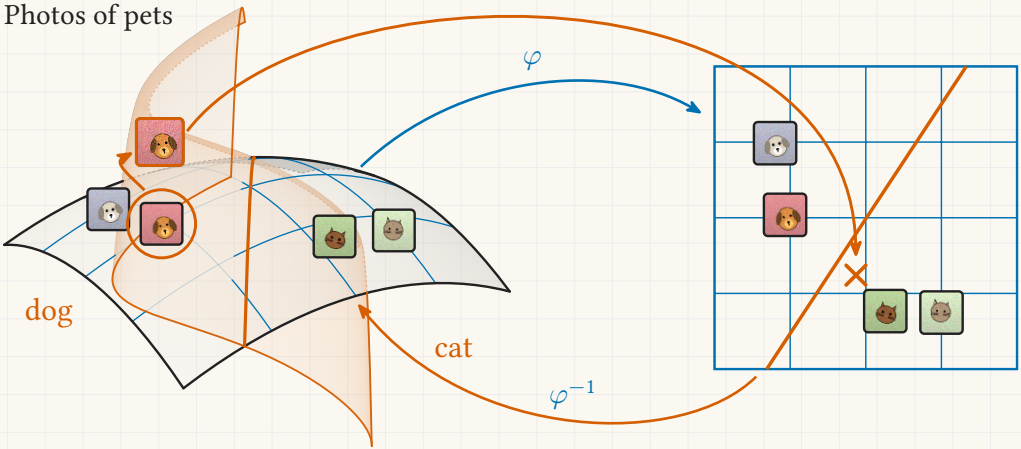
Dog or cat?


CNN
$\hat{y}$
the probability that it is a dog
$1 - \hat{y}$
the probability that it is a cat

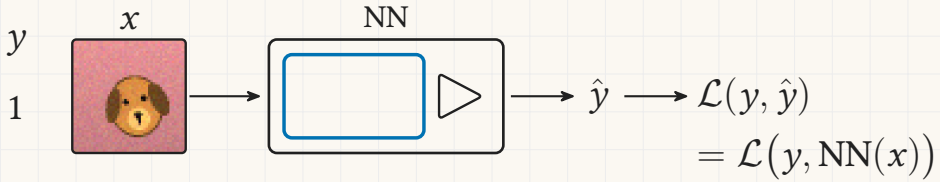
Inside the CNN



Dogs and cats on a chart



The attack



Goal: change x to change $\hat{y}$.

Add ε to x to increase $\mathcal{L}\big(y, \mathrm{NN}(x)\big)$.

$$x = x + \varepsilon \, \nabla_x \mathcal{L}\big(y, \mathrm{NN}(x)\big)$$

$$\Rightarrow \qquad x = x + \varepsilon \, \mathrm{sgn}\Big(\nabla_x \mathcal{L}\big(y, \mathrm{NN}(x)\big)\Big)$$